

LVA-008 — upstream androidx-navigation bug report (filing-ready)

Status: filing-ready bug report for the AndroidX issue tracker. **Where to file:** <https://issuetracker.google.com/issues/new?component=409828> (Component: *Android Public Tracking* → *App Development* → *Jetpack (androidx)* → *Navigation*), or the "Create issue" link on <https://developer.android.com/jetpack/androidx/releases/navigation>. Filing needs a signed-in Google account — the operator files; the body below is ready to paste.

Provenance / non-duplication note (read first)

A prior, equivalent upstream package was authored on the off-master commit **71bee48c** (`docs/issues/upstream/lva-008-androidx-navigation/UPSTREAM-ISSUE-DRAFT.md`

- `MINIMAL-REPRO.md`), held back from master for tag-time docs-sync. That package is **not present on the master working tree**, and it embeds its repro code inside markdown (there is no standalone compilable `.kt`). This directory **consolidates** the verified substance of that package and **adds** what it lacked:
- `MinimalRepro.kt` — a single, standalone, compile-correct Kotlin source file (the attachable code sample, not embedded in prose).
- `analysis.md` — the in-depth root-cause analysis with this session's falsified-fix ledger.

Nothing here was copied from speculation. Every version, route, stack frame, and falsified hypothesis is taken from in-repo device evidence (cited below). This is a consolidation/extension of an existing artifact, not a fresh duplicate claim.

Title (paste into the tracker)

NavHost: IllegalStateException "State must be at least 'CREATED' to be moved to 'DESTROYED'"
on a nested NavHost's INITIALIZED child entry at Activity destroy

Environment

Field	Value
Library	androidx.navigation:navigation-compose 2.9.1 AND 2.9.8 (latest stable, 2026-04-22); also unfixed on 2.10.0-alpha04/alpha05 (2026-05-19)
Reproduces on	
androidx.lifecycle	2.9.1 — LifecycleRegistry is what throws
Compose BOM	2025.06.01
Kotlin	2.1.0 · AGP 8.6.1
compileSdk / targetSdk / minSdk	35 / 35 / 21
Device	Genymotion Pixel 9, Android 15 / API 35, arm64-v8a (serial 127.0.0.1:6555); also reproduced on a containerized x86_64 API-34 emulator
Reproducibility	Consistent on cold-boot device runs — 3× ISE per teardown in the JUnit XML, on both affected flows (C06 download, C11 search)

Summary

Single-Activity Compose app using the **bottom-navigation-with-nested-graphs** pattern. An **inner NavHost is composed from inside a parent NavBackStackEntry** of the outer graph — so that parent entry, **not the Activity**, is the inner NavController's host LifecycleOwner.

The user navigates the inner host to a child route (in our app, search/search_input?query=...) and then the **Activity is destroyed** (rotation / "Don't keep activities" / process death / config change). During teardown the framework crashes:

```
java.lang.IllegalStateException: State must be at least 'CREATED' to be moved to 'DESTROYED',  
    but was 'INITIALIZED' in component NavBackStackEntry(...  
route=search_input ...)
```

```
at
androidx.lifecycle.LifecycleRegistry.checkLifecycleStateTransition()
```

The exception is rethrown as `RuntimeException: Unable to destroy activity inside ActivityThread.performDestroyActivity and crashes the process — it is not catchable by any in-process handler.`

Exact production stack (verbatim from device evidence)

```
Process crashed – Unable to destroy activity MainActivity
Caused by: java.lang.IllegalStateException:
    State must be at least 'CREATED' to be moved to 'DESTROYED',
but was 'INITIALIZED'
    in component NavBackStackEntry(route=search/search_input?
query={query}&categories={categories}
    &author_name={author_name}&author_id={author_id}
&sort={sort}&order={order}&period={period})
    at
androidx.lifecycle.LifecycleRegistry.checkLifecycleStateTransition()
    at
androidx.lifecycle.LifecycleRegistry.moveToState(LifecycleRegistry.
    at androidx.navigation.NavBackStackEntry... updateState
    at androidx.navigation.internal.NavControllerImpl ...
lifecycleObserver // host ON_DESTROY observer
```

Why it happens

A `NavBackStackEntry`'s effective state is `min(hostLifecycleState, maxLifecycle)` (`DESTROYED(0) < INITIALIZED(1) < CREATED(2) < STARTED(3) < RESUMED(4)`). In a nested host, a child `search_input` entry can be **tracked by the inner `NavController` while still `INITIALIZED`** (added to the back-stack queue / transitioning / restored, but never driven up to `CREATED`). At Activity destroy, the parent `NavBackStackEntry` (the inner host's `LifecycleOwner`) collapses **directly to `DESTROYED`**, and the inner controller's host-lifecycle observer attempts to move the `INITIALIZED` child straight to `DESTROYED`. `LifecycleRegistry` rejects this — **there is no lifecycle event that moves an owner down from `INITIALIZED`** (`ON_DESTROY` requires the owner to be at least `CREATED`).

This belongs to the umbrella `INITIALIZED-has-no-downward-event` family (**b/244910446**, **b/178029606**), surfaced here specifically via the **nested-host + Activity-destroy** path that strands an inner entry `INITIALIZED`.

Why it is test-environment-surfaced but a genuine lifecycle-ordering bug

We hit it in instrumentation because `ActivityScenario/connectedAndroidTest` does a **rapid create** → **drive UI** → **destroy** of `MainActivity`, which reliably destroys the Activity while the inner `search_input` entry is still `INITIALIZED`. That is a test *trigger*, not a test *artifact*:

- The stranded `search_input` entry persists `INITIALIZED` for the **whole app session** after the composed instance was popped — it is not created by the test.
- The destroy path the test exercises is the **same** path a real user crosses on rotation, dark-mode toggle, locale change, "Don't keep activities", or low-memory process death after using search. Crashlytics **confirms it live in the field** on shipped builds (FATAL; 4 events / 4 users on 1.3.11-1075).
- It reproduces on a clean cold-boot device, version-independently through 2.9.8 and into the 2.10 alpha line.

So: the test merely makes a real lifecycle-ordering defect deterministic.

Minimal reproduction

See `MinimalRepro.kt` in this directory — a complete, standalone single-Activity file with no app-specific dependencies (generic `androidx-navigation` only). Outline:

- Outer `NavHost` = bottom-nav scaffold; tab destinations are **graphs** (`navigation(route="home"){ composable("home_inner"){ InnerNavHost() } }`).
- Inner `NavHost` = inner `NavHost` (`list → search_input?query={query}`), composed from inside the home graph's `NavBackStackEntry`.

Steps: enable Developer options → "Don't keep activities" → launch → tap "Open `search_input`" (inner `list → search_input`) → press Home / rotate → **crash at destroy**. (Rotation, dark-mode, locale change, and process death trigger the same path.) See `MinimalRepro.kt`'s header KDoc for the full repro/expected/actual and the honest non-determinism note.

Expected vs actual

- **Expected:** Activity destroys cleanly; inner `search_input` state is saved and restored on re-creation.

- **Actual:**

`IllegalStateException`: State must be at least 'CREATED' to be moved to 'DESTROYED' on the inner `search_input` entry; process crash.

Workarounds tried and device-FALSIFIED (please do not re-suggest)

Every app-level / public-API avenue was implemented and then device-verified to **still crash** (or proven structurally impossible). Full evidence paths in `analysis.md`.

1. Bump nav 2.9.1 → 2.9.8 (latest stable). Identical crash.
2. Bump nav to 2.10.0-alpha04/alpha05. Identical crash.
3. `TestRule` / try-catch around teardown. **Impossible** — rethrown as `RuntimeException("Unable to destroy activity")` in `Instrumentation.callActivityOnDestroy` → `ActivityThread.performDestroyActivity`; kills the process, never propagates synchronously.
4. Move the destination to the **OUTER** `NavHost`. Crash **moved** to the outer `search_input` entry but **still fired** — disproving "nesting is strictly required"; the bug is intrinsic to an entry left `INITIALIZED`.
5. Atomic-replace nav (`popUpTo(current){inclusive=true}` + `navigate()` as one transaction). Identical crash.
6. Host `LifecycleEventObserver` that on `ON_STOP` force-advances every `INITIALIZED` entry in `navController.currentBackStack.value` to `CREATED`. Still crashes — the phantom entry is **not in the public `currentBackStack StateFlow`**; it lives in controller internals (`@RestrictTo`, reflection-only, not shipped).
7. Re-point the inner host's `LocalLifecycleOwner` to the Activity (keep `LocalViewModelStoreOwner` + `LocalSavedStateRegistryOwner` on the parent entry). Device-verified — **still crashes** identically.
8. `launchSingleTop` dedupe on the `search_input` navigation. Device-falsified (C06/C11 FAIL, `.lava-ci-evidence/lva008-cand8-gate/`).

Conclusion from (1)–(8): **no clean app-level public-API fix exists**; the defect is in `androidx-navigation`'s handling of `INITIALIZED` nested entries at host teardown.

Related / candidate-duplicate issues (all accessed 2026-06-25)

- **b/244910446** — "no event down from `INITIALIZED`" (umbrella family) — <https://issuetracker.google.com/issues/244910446>

- **b/178029606** — "no event down from INITIALIZED" (sibling) — <https://issuetracker.google.com/issues/178029606>
- **google/accompanist#1487** — NavBackStackEntry ViewModels after destroy — <https://github.com/google/accompanist/issues/1487>
- **android/compose-samples#664** — Jetsnack bottom-nav fast-switch crash — <https://github.com/android/compose-samples/issues/664>

This report adds value by isolating the **nested-host + Activity-destroy** trigger and providing a generic minimal repro that strands an inner entry INITIALIZED deterministically.

The ask

1. **Confirm** whether NavControllerImpl's host-lifecycle observer should no-op (or first drive up to CREATED) when asked to move an INITIALIZED nested entry to DESTROYED, rather than calling LifecycleRegistry with an illegal INITIALIZED → DESTROYED transition.
2. **Fix** so destroying the Activity / parent NavBackStackEntry does not crash when an inner NavHost has an INITIALIZED child — destroy it cleanly the way CREATED+ entries are.
3. If this is a **dup of b/244910446**, link it and indicate the **target fix version** — release notes do not list a fix through 2.9.8 or 2.10.0-alpha05.

In-repo evidence paths (Lava-internal; sanitize/optional when filing)

- Incident JSON (verbatim crash signature + falsified-hypothesis ledger):
`.lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json`
- Device-gate JUnit XML showing 3× IllegalStateException per teardown: `.lava-ci-evidence/1074-gate/ (C06, C11)` and `.lava-ci-evidence/lva008-cand7-gate/`, `.lava-ci-evidence/lva008-cand8-gate/`
- Latest baseline runs: `.lava-ci-evidence/genymotion/full-baseline-runnerfix-20260626/`, `.lava-ci-evidence/genymotion/hyp-nav298-c05/`
- 1076 reproduction (C52 search chips + C48 sync toggle both crash identically): `.lava-ci-evidence/1076-repro/`
- §6.AK incident (C00-only gate shipped broken flows): `.lava-ci-evidence/sixth-law-incidents/2026-06-26-c00-only-gate-shipped-broken-flows.json`
- Bugfix log: `docs/issues/fixed/BUGFIXES.md` (LVA-008 entry)

- Prior off-master upstream package: commit 71bee48c, docs/issues/upstream/lva-008-androidx-navigation/{UPSTREAM-ISSUE-DRAFT,MINIMAL-REPRO}.md

Sources (all accessed 2026-06-25)

- Navigation release notes (stable 2.9.8 / 2026-04-22; latest 2.10.0-alpha05 / 2026-05-19): <https://developer.android.com/jetpack/androidx/releases/navigation>
- AndroidX versions index: <https://developer.android.com/jetpack/androidx/versions>
- navigation-compose on Maven Central: <https://mvnrepository.com/artifact/androidx.navigation/navigation-compose>
- b/244910446: <https://issuetracker.google.com/issues/244910446>
- b/178029606: <https://issuetracker.google.com/issues/178029606>
- accompanist#1487: <https://github.com/google/accompanist/issues/1487>
- compose-samples#664: <https://github.com/android/compose-samples/issues/664>
- NavBackStackEntry lifecycle explainer (effectiveState = min(hostLifecycleState, maxLifecycle)): <https://medium.com/@anteprocess/android-navbackstackentrys-lifecycle-in-jetpack-compose-navigation-d596fcc29fc4>